

# Karthik Bhattaram

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## EDUCATION

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### University of California, Santa Barbara

Santa Barbara

*Master of Science in Computer Science*

*March 2026 to June. 2027*

- Courses (planned): Distributed Systems, Advanced Topics in Security, Database Systems
- Research Area: Network Security, supervised by Chris Kruegel and Giovanni Vigna

### University of California, Santa Barbara

Santa Barbara

*Bachelor of Science in Computer Science*

*Sept. 2022 to Mar. 2026*

- Courses: Machine Learning for Networking, Computer Security, Data Structures and Algorithms
- Honors: Regents Scholar, College of Engineering Honors, USACO Silver Divison

## EXPERIENCE

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### Software Engineer Intern

June 2025 – Sept. 2025

*Arista Networks*

*Santa Clara, California*

- Accelerated network switch memory diagnostics by orders of magnitude using custom C++ utilities
- Developed an **interprocess communication** tool to relay this information and save hundreds of gigabytes of data

### Machine Learning Research Intern

Sept. 2024 – June 2025

*University of California, Santa Barbara*

*Santa Barbara, California*

- Published a paper titled “Leveraging Large Language Models to Contextualize Network Measurements”
- Aggregated FCC data into a SQL database to improve model context using **retrieval-augmented generation**

### Software Engineer Intern

July 2024 – Sept. 2024

*Cadence Design Systems / OpenEye Scientific*

*Santa Fe, New Mexico*

- Wrote **AWS**-based package to benchmark database compression algorithms that saved terabytes of server space
- Updated **CUDA** C++ software to incorporate new language features to match competing industry solutions

### Hardware Architecture Intern

June 2023 – Sept. 2023

*Cadence Design Systems*

*San Jose, California*

- Implemented an automation tool that produced computer memories to accelerate chip design
- Collected **System Verilog** files to be used as templates for this tool and tested memories it produced

### Software Engineer Intern

June 2021 – Aug. 2022

*Axiado Corporation*

*San Jose, California*

- Developed a package to create **network packet flows** based on IPv6, ARP, ICMP, etc. in simulation
- Created and executed a test plan for a hardware firewall’s parser module and discovered multiple bugs

## PROJECTS

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### College Course Database | *Java, SQL*

March 2025 – May 2025

- Designed a database schema using Entity-Relationship diagrams
- Wrote SQL commands to create the tables and enforce data integrity
- Implemented a Java backend with JDBC to manage database transactions
- Created command line interfaces for the registrar and students with distinct permissions

## TECHNICAL SKILLS

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**Certificates:** Enterprise Security Professional, Data Science and AI

**Networking Technologies:** Scapy, Wireshark, TCP/IP, Ethernet

**Languages:** Python, C/C++, Java, JavaScript, Shell, Verilog

**Developer Tools:** Git, Visual Studio Code, Vim, Emacs

**Libraries:** CUDA, Pandas, NumPy, PyTest, Black, Matplotlib